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<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Malaysian Agrotech Crop Growth Derivative Simulator</title>
  <!-- Tailwind CSS CDN -->
  <script src="https://cdn.tailwindcss.com"></script>
  <!-- Chart.js CDN -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <!-- FontAwesome Icons -->
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.4.0/css/all.min.css">
  <!-- MathJax for rendering LaTeX formulas beautifully -->
  <script src="https://polyfill.io/v3/polyfill.min.js?features=es6"></script>
  <script id="MathJax-script" async src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-mml-chtml.js"></script>
  <style>
    @import
url('https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@300;400;500;600;700;800&display=swap');
    body {
      font-family: 'Plus Jakarta Sans', sans-serif;
    }
  </style>
</head>
<body class="bg-slate-50 text-slate-800 min-h-screen">

  <!-- Header / Navbar -->
  <header class="bg-emerald-800 text-white shadow-md sticky top-0 z-50">
    <div class="max-w-7xl mx-auto px-4 py-4 sm:px-6 lg:px-8 flex flex-col sm:flex-row justify-between items-center gap-4">
      <div class="flex items-center space-x-3">
        <div class="bg-white p-2.5 rounded-xl shadow-inner text-emerald-800 flex items-center justify-center">
          <i class="fa-solid fa-seedling text-2xl"></i>
        </div>
        <div>
          <h1 class="text-xl font-bold tracking-tight">Agrotech Bioscience
Portal</h1>
          <p class="text-xs text-emerald-200">Interactive Mathematical Derivative
Analysis</p>
        </div>
      </div>
      <div class="flex items-center space-x-2 bg-emerald-700/50 px-4 py-2 rounded-lg border border-emerald-600/30">
        <span class="text-xs font-semibold text-emerald-100 bg-emerald-900 px-2 py-1 rounded">SFA: AGROTECH</span>
        <span class="text-xs text-emerald-100">Academic Simulator</span>
      </div>
    </div>
  </header>

  <!-- Main Content Grid -->
  <main class="max-w-7xl mx-auto px-4 py-8 sm:px-6 lg:px-8">

    <!-- Scenario Introduction -->
    <section class="bg-white rounded-2xl shadow-sm border border-slate-100 p-6 mb-8">
      <div class="flex flex-col lg:flex-row gap-8 items-start">
        <div class="lg:w-2/3">

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<div class="inline-flex items-center gap-2 bg-emerald-50 text-emerald-800
px-3 py-1 rounded-full text-xs font-semibold mb-3">
  <span class="w-2 h-2 rounded-full bg-emerald-500 animate-
pulse"></span>

  Malaysian Smart Farming Focus
</div>
<h2 class="text-2xl font-bold text-slate-950 mb-4">
  Exponential Modeling of Fertilized Chilli (<span class="italic text-
emerald-700">Cili Kulai</span>) Growth
</h2>
<p class="text-slate-600 leading-relaxed mb-4 text-sm sm:text-base">
  In Malaysian agrotechnology, optimizing the growth cycle of modern
crop varieties like

  <strong>Cili Kulai</strong> in smart fertigation greenhouses is
essential for food security and export premium.
  To mathematically track dry shoot biomass ($W$ in grams) over time
($t$ in days after transplanting), agronomists apply the
  <strong>Exponential Growth Model</strong>. This model handles the
biological transition from early exponential growth
  (where cells divide unimpeded) to linear growth (governed by canopy
sunlight absorption limits).
</p>
<p class="text-slate-600 leading-relaxed text-sm sm:text-base">
  The <strong>derivative $f'(t) = \frac{dW}{dt}$</strong> is the
critical agronomic indicator. It represents the
  <strong>instantaneous growth rate (g/day)</strong>. Pinpointing the
maximum derivative value and monitoring how
  this rate slows as the plant matures allows agronomists to optimize
fertilization schedules (NPK ratios) and predict precise harvests.
</p>
</div>
<!-- Interactive Summary Card -->
<div class="lg:w-1/3 w-full bg-slate-50 rounded-xl border border-slate-200/60
p-5">

  <h3 class="font-bold text-slate-900 text-sm uppercase tracking-wider mb-3
flex items-center gap-2 text-emerald-800">
    <i class="fa-solid fa-database"></i> Realistic Dataset References
  </h3>
  <ul class="space-y-3 text-xs text-slate-600">
    <li class="flex items-start gap-2">
      <i class="fa-solid fa-bookmark text-emerald-600 mt-0.5"></i>
      <div>
        <span class="font-semibold text-slate-900 block">MARDI Crop
Database</span>

        Growth profiles for modern Malaysian chilli variants.
      </div>
    </li>
    <li class="flex items-start gap-2">
      <i class="fa-solid fa-bookmark text-emerald-600 mt-0.5"></i>
      <div>
        <span class="font-semibold text-slate-900 block">Goudriaan &
Monteith model (1990)</span>

        Standard framework for exponential dry biomass development.
      </div>
    </li>
    <li class="flex items-start gap-2">
      <i class="fa-solid fa-globe text-emerald-600 mt-0.5"></i>
      <div class="break-all">
        <span class="font-semibold text-slate-900 block">Source
webpage:</span>

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                <a href="https://pmc.ncbi.nlm.nih.gov/" target="_blank"
class="text-emerald-700 hover:underline font-medium">PMC: Expolinear Crop Equations</a>
            </div>
        </li>
    </ul>
</div>
</div>
</section>

<!-- Math Formulas Panel -->
<section class="grid grid-cols-1 md:grid-cols-2 gap-6 mb-8">
    <!-- Growth Function f(t) -->
    <div class="bg-emerald-950 text-white rounded-2xl p-6 shadow-sm border border-emerald-800">
        <div class="flex justify-between items-center mb-4">
            <span class="text-xs uppercase tracking-widest text-emerald-400 font-bold">Crop Biomass Equation</span>
            <span class="bg-emerald-800 text-emerald-100 text-xs px-2.5 py-1 rounded-md font-medium">Function f(t)</span>
        </div>
        <h3 class="text-lg font-bold mb-3">Dry Shoot Weight: $W(t)$</h3>
        <div class="bg-emerald-900/60 p-4 rounded-xl border border-emerald-800 flex justify-center items-center my-3 overflow-x-auto">
            <p class="text-lg font-mono">
                $$W(t) = \frac{C_m}{r_m} \ln\left[1 + e^{r_m(t - t_b)}\right]$$
            </p>
        </div>
        <p class="text-xs text-emerald-200 leading-relaxed">
            Where:<br>
            • $C_m = 6.8 \text{ g/day}$: Maximum absolute growth rate under Malaysian solar radiation.<br>
            • $r_m = 0.18 \text{ day}^{-1}$: Maximum relative growth rate in early vegetative phase.<br>
            • $t_b = 18 \text{ days}$: The "lost time" parameter representing crop establishment delay.
        </p>
    </div>

    <!-- Derivative Function f'(t) -->
    <div class="bg-indigo-950 text-white rounded-2xl p-6 shadow-sm border border-indigo-900">
        <div class="flex justify-between items-center mb-4">
            <span class="text-xs uppercase tracking-widest text-indigo-400 font-bold">Growth Velocity Equation</span>
            <span class="bg-indigo-800 text-indigo-100 text-xs px-2.5 py-1 rounded-md font-medium">Derivative f'(t)</span>
        </div>
        <h3 class="text-lg font-bold mb-3">Instantaneous Growth Rate: $W'(t)$</h3>
        <div class="bg-indigo-900/60 p-4 rounded-xl border border-indigo-800 flex justify-center items-center my-3 overflow-x-auto">
            <p class="text-lg font-mono">
                $$W'(t) = \frac{dW}{dt} = C_m \cdot \frac{e^{r_m(t-t_b)}}{1 + e^{r_m(t-t_b)}}$$
            </p>
        </div>
        <p class="text-xs text-indigo-200 leading-relaxed">
            Derived using the Chain Rule:<br>
            Let $u = 1 + e^{r_m(t-t_b)}$, then $W = \frac{C_m}{r_m} \ln(u)$.<br>
            $\frac{dW}{dt} = \frac{dW}{du} \cdot \frac{du}{dt} = \left(\frac{C_m}{r_m} \cdot \frac{1}{u}\right) \cdot \left(r_m e^{r_m(t-t_b)}\right)$, which simplifies directly to the velocity formula.
        </p>
    </div>

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        </p>
    </div>
</section>

<!-- Main Chart and Interactive Board -->
<section class="bg-white rounded-2xl shadow-sm border border-slate-100 p-6 mb-8">
    <div class="flex justify-between items-center mb-6">
        <div>
            <h3 class="text-lg font-bold text-slate-900">Chilli Canopy Growth Curve &
Instantaneous Rate</h3>
            <p class="text-xs text-slate-500">Interactive simulator: Click anywhere on
the graph to evaluate the derivative at that exact day.</p>
        </div>
        <div class="flex gap-2">
            <button id="btnPlay" class="bg-emerald-600 hover:bg-emerald-700 text-white
px-4 py-1.5 rounded-lg text-xs font-semibold flex items-center gap-1 transition shadow-sm">
                <i class="fa-solid fa-play"></i> Animate Growth
            </button>
            <button id="btnReset" class="bg-slate-100 hover:bg-slate-200 text-slate-
700 px-3 py-1.5 rounded-lg text-xs font-semibold transition border border-slate-200">
                Reset
            </button>
        </div>
    </div>
</div>

<!-- Responsive Chart Canvas Container -->
<div class="relative bg-slate-50 rounded-xl p-4 mb-6 border border-slate-100
overflow-hidden" style="height: 400px;">
    <canvas id="growthChart" class="w-full h-full"></canvas>
</div>

<!-- Real-time Derivative Evaluation Panel -->
<div id="interactionPanel" class="bg-emerald-50/50 border border-emerald-100
rounded-xl p-6 transition-all duration-300">
    <div class="flex flex-col md:flex-row justify-between items-start md:items-
center gap-6">
        <div class="flex items-center gap-4">
            <div class="bg-emerald-600 text-white p-4 rounded-xl shadow-md text-
center min-w-[90px]">
                <span class="block text-xs uppercase tracking-wider text-emerald-
200 font-bold mb-0.5">Selected</span>
                <span id="displayDay" class="text-2xl font-black">Day 25</span>
            </div>
            <div>
                <span class="text-xs font-bold text-slate-400 uppercase tracking-
widest block">Simulation Evaluation Point</span>
                <h4 id="displayPhase" class="text-lg font-bold text-slate-
900">Linear Canopy Cover Expansion Phase</h4>
            </div>
        </div>
        <div class="grid grid-cols-2 gap-4 w-full md:w-auto">
            <div class="bg-white p-3 rounded-lg border border-slate-100 shadow-sm
min-w-[150px]">
                <span class="text-xs text-slate-500 block">Dry Biomass
                $f(t)$</span>
                <span id="displayWeight" class="text-xl font-bold text-emerald-
800">54.0 g</span>
            </div>
            <div class="bg-white p-3 rounded-lg border border-slate-100 shadow-sm
min-w-[150px]">

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        <span class="text-xs text-slate-500 block">Growth Rate
    $f'(t)$</span>
        <span id="displayDerivative" class="text-xl font-bold text-indigo-
800">5.34 g/day</span>
    </div>
</div>
</div>

<!-- Layman Explanation Box -->
<div class="mt-5 pt-4 border-t border-emerald-100">
    <div class="flex gap-2.5 items-start">
        <div class="bg-emerald-100 text-emerald-800 p-1.5 rounded-lg mt-0.5">
            <i class="fa-solid fa-graduation-cap"></i>
        </div>
        <div>
            <span class="text-xs font-bold text-emerald-800 uppercase
tracking-wider block mb-1">Agronomist Interpretation</span>
            <p id="laymanExplanation" class="text-slate-700 text-sm leading-
relaxed">
                At day 25, the chilli seedling has fully completed its nursery
recovery phase. Its canopy is closing rapidly, intercepts optimal daylight, and its biomass
increases by 5.34 grams per day. This is a critical stage to ensure proper fertigation EC
(electrical conductivity) levels to sustain high-rate growth.
            </p>
        </div>
    </div>
</div>
</div>
</div>
</section>

<!-- Dynamic Agronomic Slider for Manual Inspection -->
<section class="bg-white rounded-2xl shadow-sm border border-slate-100 p-6 mb-8">
    <h3 class="font-bold text-slate-900 mb-4 flex items-center gap-2">
        <i class="fa-solid fa-sliders text-emerald-700"></i> Scrub Growth Timeline
    </h3>
    <div class="space-y-4">
        <div class="flex justify-between text-xs text-slate-500 font-semibold">
            <span>Transplanting (Day 0)</span>
            <span class="bg-indigo-100 text-indigo-800 px-2 py-0.5 rounded">Active
Focus: <span id="sliderLabel">Day 25</span></span>
            <span>Mature Harvesting (Day 50)</span>
        </div>
        <input id="timelineSlider" type="range" min="0" max="50" value="25" class="w-
full h-2 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-emerald-600">
    </div>
</section>

<!-- Educational Section explaining the Bio-Math Connection -->
<section class="bg-white rounded-2xl shadow-sm border border-slate-100 p-6">
    <h3 class="text-xl font-bold text-slate-900 mb-4">Understanding the Bioscience of
the Derivative</h3>
    <div class="grid grid-cols-1 md:grid-cols-3 gap-6 text-sm">
        <div class="border-l-4 border-emerald-600 pl-4 py-1">
            <h4 class="font-bold text-slate-950 mb-1">Stage 1: Exponential Phase ($t <
15$ days)</h4>
            <p class="text-slate-600 leading-relaxed text-xs">
                The plant is small and doesn't shade its own leaves. Every new cell
increases photosynthetic potential, driving exponential speedups. The derivative $f'(t)$ is
low but accelerating upwards rapidly.
            </p>
        </div>
    </div>

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        <div class="border-l-4 border-amber-500 pl-4 py-1">
            <h4 class="font-bold text-slate-950 mb-1">Stage 2: Linear Phase ($15 \le t
\le 30$ days)</h4>
            <p class="text-slate-600 leading-relaxed text-xs">
                The canopy closes, fully intercepting incident light. The absolute
                biomass growth rate peaks and hits a flat ceiling ( $C_m = 6.8 \text{ g/day}$ ). Here, the
                derivative remains consistently high and nearly constant.
            </p>
        </div>
        <div class="border-l-4 border-slate-400 pl-4 py-1">
            <h4 class="font-bold text-slate-950 mb-1">Stage 3: Saturation Phase ($t >
30$ days)</h4>
            <p class="text-slate-600 leading-relaxed text-xs">
                Lower leaves become shaded, and energy goes towards maintaining
                structural stem mass rather than creating brand-new leaves. The derivative slope flattens,
                indicating growth velocity slowing as harvesting approaches.
            </p>
        </div>
    </div>
</section>

</main>

<!-- Footer -->
<footer class="bg-slate-900 text-slate-400 py-12 mt-16 border-t border-slate-800 text-xs">
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 text-center space-y-4">
        <p>Designed for undergraduate students in Malaysian agricultural technologies (SFA:
        Agrotech Programs).</p>
        <p class="text-slate-500">
            Mathematics modeled on the classic crop growth differential templates by
            Goudriaan, J., and Monteith, J. L. (1990) "A mathematical function for crop growth", Ann. Bot.
            66: 695-701.
        </p>
        <p>&copy; 2026 Malaysia Agrotech Academic Resources. All rights reserved.</p>
    </div>
</footer>

<!-- Logic Script for Calculations, Animation, Chart.js Initialization -->
<script>
    // Parameters for the Exponential Chilli Growth Model
    const Cm = 6.8; // Max linear growth rate (grams/day)
    const rm = 0.18; // Max relative growth rate (1/day)
    const tb = 18; // Time-lag offset (days)
    const maxDays = 50;

    // Mathematical formulation of Exponential growth f(t)
    function f(t) {
        return (Cm / rm) * Math.log(1 + Math.exp(rm * (t - tb)));
    }

    // Mathematical formulation of its derivative f'(t)
    function df(t) {
        const expPart = Math.exp(rm * (t - tb));
        return Cm * (expPart / (1 + expPart));
    }

    // Generate datasets for the chart
    const daysLabel = [];
    const biomassData = [];
    const growthRateData = [];

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for (let t = 0; t <= maxDays; t += 0.5) {
  daysLabel.push(t);
  biomassData.push(f(t));
  growthRateData.push(df(t));
}

// Select DOM Elements
const timelineSlider = document.getElementById('timelineSlider');
const sliderLabel = document.getElementById('sliderLabel');
const displayDay = document.getElementById('displayDay');
const displayPhase = document.getElementById('displayPhase');
const displayWeight = document.getElementById('displayWeight');
const displayDerivative = document.getElementById('displayDerivative');
const laymanExplanation = document.getElementById('laymanExplanation');
const btnPlay = document.getElementById('btnPlay');
const btnReset = document.getElementById('btnReset');

// Chart.js Configuration
const ctx = document.getElementById('growthChart').getContext('2d');
let activeXValue = 25; // Default focus point on Day 25

// Vertical line plugin to highlight the evaluated day
const verticalLinePlugin = {
  id: 'verticalLine',
  afterDraw: (chart) => {
    if (activeXValue === null) return;
    const { ctx, chartArea: { top, bottom }, scales: { x } } = chart;
    const xCoord = x.getPixelForValue(activeXValue);

    ctx.save();
    ctx.beginPath();
    ctx.setLineDash([6, 4]);
    ctx.moveTo(xCoord, top);
    ctx.lineTo(xCoord, bottom);
    ctx.lineWidth = 2;
    ctx.strokeStyle = '#4f46e5'; // Indigo color matching active day
    ctx.stroke();
    ctx.restore();
  }
};

const chart = new Chart(ctx, {
  type: 'line',
  data: {
    labels: daysLabel,
    datasets: [
      {
        label: 'Dry Crop Weight f(t) (grams)',
        data: biomassData,
        borderColor: '#059669', // Emerald
        backgroundColor: 'rgba(5, 150, 105, 0.05)',
        borderWidth: 3.5,
        fill: true,
        tension: 0.1,
        yAxisID: 'y'
      },
      {
        label: "Instantaneous Growth Rate f'(t) (g/day)",
        data: growthRateData,
        borderColor: '#6366f1', // Indigo
        borderWidth: 2,

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        borderDash: [5, 5],
        fill: false,
        yAxisID: 'yDerivative'
    }
]
},
options: {
    responsive: true,
    maintainAspectRatio: false,
    interaction: {
        mode: 'index',
        intersect: false,
    },
    plugins: {
        legend: {
            position: 'top',
            labels: {
                font: {
                    family: 'Plus Jakarta Sans',
                    weight: 'bold',
                    size: 11
                }
            }
        },
        tooltip: {
            enabled: true,
            callbacks: {
                title: function(context) {
                    return `Day ${parseFloat(context[0].label).toFixed(1)}`;
                }
            }
        }
    },
    scales: {
        x: {
            type: 'linear',
            min: 0,
            max: maxDays,
            title: {
                display: true,
                text: 'Days After Transplanting (t)',
                color: '#1e293b',
                font: {
                    family: 'Plus Jakarta Sans',
                    weight: 'bold'
                }
            },
            grid: {
                color: 'rgba(241, 245, 249, 1)'
            }
        },
        y: {
            type: 'linear',
            position: 'left',
            title: {
                display: true,
                text: 'Dry Shoot Biomass (g)',
                color: '#059669',
                font: {
                    family: 'Plus Jakarta Sans',
                    weight: 'bold'
                }
            }
        }
    }
}

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        }
      },
      suggestedMax: 200,
      grid: {
        color: 'rgba(241, 245, 249, 1)'
      }
    },
    yDerivative: {
      type: 'linear',
      position: 'right',
      title: {
        display: true,
        text: 'Instantaneous Growth Velocity (g/day)',
        color: '#6366f1',
        font: {
          family: 'Plus Jakarta Sans',
          weight: 'bold'
        }
      }
    },
    min: 0,
    max: 8,
    grid: {
      drawOnChartArea: false // only want grid lines for main y-axis
    }
  },
  onClick: (e) => {
    const points = chart.getElementsAtEventForMode(e, 'nearest', { intersect:
false }, true);
    if (points.length) {
      const index = points[0].index;
      const clickedDay = daysLabel[index];
      updateSimulation(clickedDay);
    }
  },
  plugins: [verticalLinePlugin]
});

// Returns dynamic layman agronomic explanations tailored for Malaysian chilli
agrotech context
function getLaymanExplanation(day, weight, rate) {
  if (day < 10) {
    return `At day ${day.toFixed(1)}, the Cili Kulai seedling is establishing root
networks in the fertigation coco-peat bags. Biomass is low (${weight.toFixed(1)}g), and growth
is slow but beginning to speed up. The rate of change is a minor ${rate.toFixed(2)}g/day as it
recovers from transplant shock. Ensure high phosphorus supply for root development now!`;
  } else if (day >= 10 && day < 20) {
    return `Between days 10 and 20, the chilli is in its exponential vegetative
phase. Every day, new leaves emerge to capture solar radiation. Today (Day ${day.toFixed(1)}),
the crop's daily weight gain is accelerating up to ${rate.toFixed(2)}g/day. This represents
the explosive phase of biological compound interest before shading limits kick in.`;
  } else if (day >= 20 && day < 32) {
    return `Now at day ${day.toFixed(1)}, the crop is entering its peak physical
performance phase. The canopy has closed, meaning there is maximum sunlight interception in
your greenhouse. The growth rate is stable near its biological maximum of
${rate.toFixed(2)}g/day. Fertigation nutrients (especially high Nitrogen and Potassium) must
be strictly sustained to support this heavy cellular synthesis.`;
  } else {
    return `At Day ${day.toFixed(1)}, the chilli crop has entered its reproductive
and saturation phase. Heavy biomass is now diverted into chilli pod development (flowering and

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fruiting). The instantaneous vegetative growth velocity has slowed down to
 $\text{\${rate.toFixed(2)}}\text{g/day}$  because the plant is maintaining existing structures rather than
expanding new canopy. This signals readiness for transition to fruit-harvesting maintenance
nutrients.`;
    }
}

// Returns descriptive phase text
function getGrowthPhase(day) {
    if (day < 10) return "Seedling & Root Development Phase";
    if (day >= 10 && day < 20) return "Exponential Vegetative Acceleration Phase";
    if (day >= 20 && day < 32) return "Peak Canopy Growth & Fertilization Max Efficiency
Phase";
    return "Reproductive, Pod Development & Saturation Phase";
}

// Synchronize and update the UI
function updateSimulation(day) {
    day = Math.min(Math.max(0, day), maxDays);
    activeXValue = parseFloat(day.toFixed(1));

    // Calculate values using formulas
    const weight = f(day);
    const rate = df(day);

    // Update DOM
    sliderLabel.textContent = `Day  $\text{\${activeXValue}}$ `;
    timelineSlider.value = activeXValue;
    displayDay.textContent = `Day  $\text{\${activeXValue}}$ `;
    displayPhase.textContent = getGrowthPhase(activeXValue);
    displayWeight.textContent = ` $\text{\${weight.toFixed(1)}}\text{ g}$ `;
    displayDerivative.textContent = ` $\text{\${rate.toFixed(2)}}\text{ g/day}$ `;
    laymanExplanation.textContent = getLaymanExplanation(activeXValue, weight, rate);

    // Redraw chart to update vertical guideline location
    chart.update();
}

// Slider Interaction
timelineSlider.addEventListener('input', (e) => {
    updateSimulation(parseFloat(e.target.value));
});

// Animation Loop Variable
let animationInterval = null;

function startAnimation() {
    clearInterval(animationInterval);
    let currentDay = 0;
    btnPlay.disabled = true;
    btnPlay.classList.add('opacity-50');

    animationInterval = setInterval(() => {
        if (currentDay > maxDays) {
            clearInterval(animationInterval);
            btnPlay.disabled = false;
            btnPlay.classList.remove('opacity-50');
            return;
        }
        updateSimulation(currentDay);
        currentDay += 1;
    }, 1000);
}

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    }, 100);  
  }  
  
  btnPlay.addEventListener('click', () => {  
    startAnimation();  
  });  
  
  btnReset.addEventListener('click', () => {  
    clearInterval(animationInterval);  
    btnPlay.disabled = false;  
    btnPlay.classList.remove('opacity-50');  
    updateSimulation(25); // reset back to middle default Day 25  
  });  
  
  // Initialize default UI values on load  
  window.onload = function() {  
    updateSimulation(25);  
  };  
</script>  
</body>  
</html>
```